Victor Jean Viaunel

Full-stack developer and civil engineer · BIM, reality capture and applied AI

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I build software that engineers and families actually use. For two years I've been the lead developer of a live tutoring platform, from design to production, and at NTU, where I'm an M.S. student and assistant researcher, I research how to turn LiDAR scans into BIM models.

EXPERIENCE

Jun 2024 – Present — **Lead Full-Stack Developer** · Legend Castle, Taiwan

- Main developer of MaiSchool, a live online tutoring platform for families, from first design to production.
- Built course and teacher browsing, class booking around family schedules, progress tracking and parent messaging.

Next.js · Convex · Clerk · Zod

Oct 2024 – Present — **Assistant Researcher** · H-Team lab, National Taiwan University

- Research on scan-to-BIM with mobile LiDAR and LLM-assisted BIM modelling; first-author papers at ISARC 2026 and ICCCBE 2026, co-author at i3CE 2026.
- Teaching assistant for Prof. Shang-Hsien Hsieh's BIM course, supporting students with practical topics and implementation ideas.

Feb – Jun 2023 — **Structural Engineering Intern** · Envision

Assistant structural engineer; structural modelling and analysis in ETABS and SAP2000.

Jun – Aug 2022 — **Research Intern, Computer-Aided Engineering** · National Taiwan University

Computer vision in OpenCV and PyTorch; sequence models with LSTMs and GRUs.

SELECTED PROJECTS

2026 — **Marine debris detection** · Taiwan International AI Challenge

YOLO detector for 34 kinds of ocean waste, two-person team: 15th of 69 in the preliminary round (mAP@0.5 0.7917). My error analysis traced most misses to ambiguous class definitions.

2026 — **Solar panel defect detection** · AI CUP, Ministry of Education

Raised the public score from 0.301 to 0.8337, finishing about 29th of 121 teams.

In progress — **PhotoForge** · offline photogrammetry with SuperPoint, LightGlue, Depth Anything V2 and SAM2

In progress — **Haitian Creole dataset** · traceable LLM training data from 42 curated sources

EDUCATION

2024 – 2027 — **M.S. Civil Engineering** · National Taiwan University · in progress, expected Jan 2027 · scan-to-BIM with mobile LiDAR

2019 – 2023 — **B.S. Civil Engineering** · National Yang Ming Chiao Tung University

PUBLICATIONS

2026 — **First-author:** *From 2D Drawings to BIM Models: An LLM-Assisted Approach* (ISARC 2026, pp. 2410–2414, doi:10.22260/ISARC2026/0307); *A Review on Scan-to-BIM for Building Energy Retrofit* (ICCCBE 2026).

LEADERSHIP

Current · 2nd term — **Academic Committee Lead** · Haitian student community in Taiwan

Guide bachelor's to graduate applicants through Taiwanese university admissions and coordinate application workshops.

SKILLS

Web	Next.js, React, Convex, Clerk, Zod, Django, FastAPI, Three.js
AI and data	Python, PyTorch, OpenCV, YOLO, LLM agents, n8n, MongoDB, Docker
Engineering	Revit, IFC, AutoCAD, ETABS, SAP2000, SAFE, LiDAR point clouds, photogrammetry
Languages	English (fluent), Haitian Creole (native), Chinese (intermediate listening, basic speaking)